

the satellite as a 'wave' file (.wav) but also displays the raw image while decoding, and further displays more images after processing the raw data once the satellite pass has been completed. Use the licensed freeware version only, until you are ready to purchase the full-featured licensed version. In the author's view, the basic freeware is sufficient for an individual experimenter. Refer

the screenshot in Fig. 15. Visit the website <https://wxtoimgrestored.xyz/> for downloading and details about this software.

Virtual audio cable. A point to keep in mind is that the receiver software (like SDR# or SDRUno) functions separately and demodulates the RF signal and then produces the audio signal from the FM transmission. The decoding software

(like WXtoImg) expects the audio signal to be fed for the recording and processing of the audio and convert it into the image.

In older days we used to connect the audio cable (called patch-cord) from the output of the receiver to the input of the recorder. Now, since our receiver is a software-defined radio, and all the demodulation and audio recovery related work take place in the software, we need a 'soft-cord' to connect the output of the receiver to the APT decoder software. For this, besides various options, the author used the VB-Cable as a freeware version. Details and software download can be found at <https://www.vb-audio.com/index.htm>

Follow the installation guide for this virtual audio cable very carefully. Once the installation is complete, check in the audio settings of your computer; the virtual cable would be visible and available for setting as shown in Fig. 17. Next, depending on your PC's sound configuration, select the Audio Out from the receiver to the Input of this cable as shown below:

Receiver Audio Output→Virtual Cable Input

Open the WXtoImg software, locate the Options menu and select the Recording option. Under this locate the Soundcard option and choose the Output from this virtual cable, or as permitted by your PC sound options.

Virtual Cable Output→WXtoImg Options→Recording→Common Recording

(EFY Lab note. During testing, the virtual cable option was not successful in Windows 10 but worked well with Windows 8.1. It is working well with default sound settings in Windows 10. You should check the sound device and settings in your PC in case it does not work.)

To test whether your audio connections are proper, follow the steps given below:

Step 1. Start your receiver software and tune to the FM stations. Keep volume level low (15 to 20)

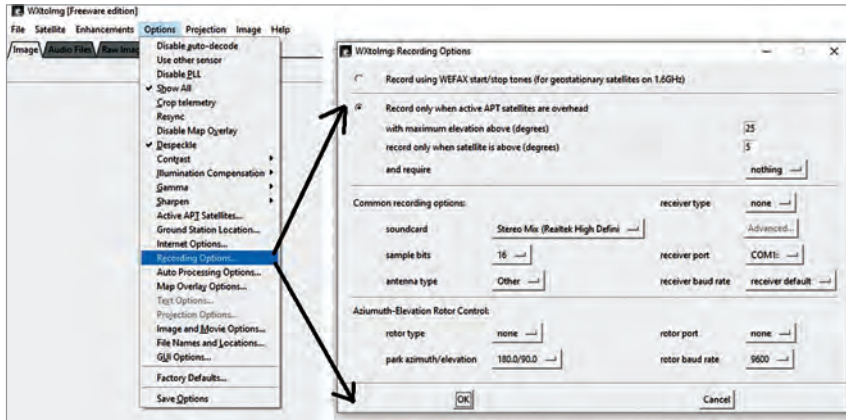


Fig. 18: Sound Setting option under the WXtoImg Decoding Software

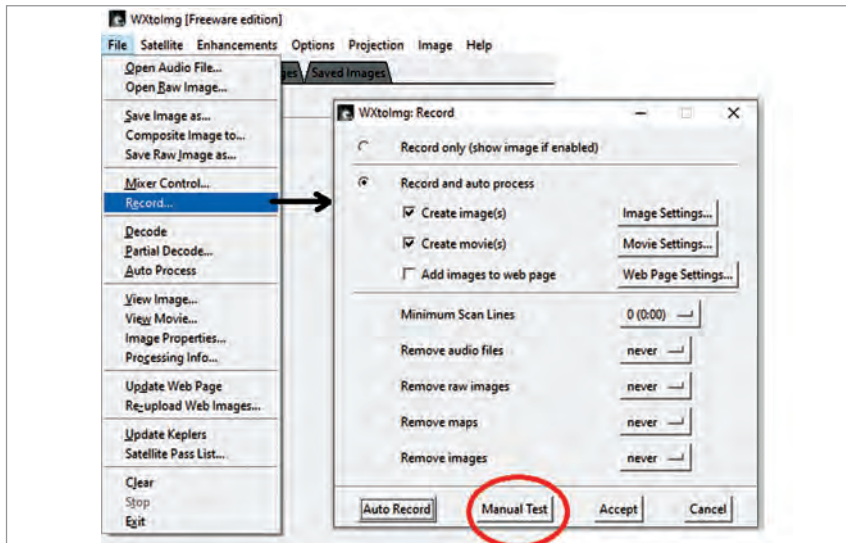


Fig. 19: Checking the sound setting using actual satellite recording menu and 'Manual' settings

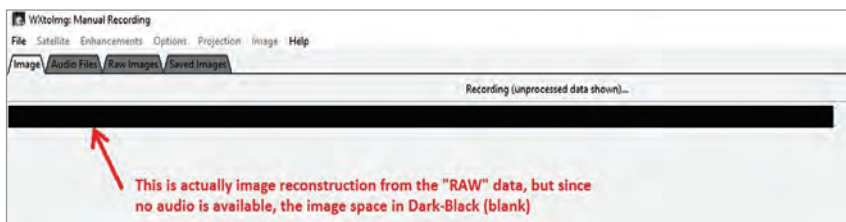


Fig. 20: Image cannot be reconstructed if audio connections are not proper